PROJECT A.R.C. (AUTONOMOUS ROBOTIC CLEANER)

Master System Architecture, Mechanical v2.0 Blueprints, Mathematical Validations, Computer Vision Pipeline, Quality Assurance Protocols, and Consolidated Procurement Manifest

Project A.R.C. Core Engineering Team

Robotics & Autonomous Systems · Techfest Innovation Challenge

Master Engineering Specification | October 2026

Abstract

This document establishes the unified, verified engineering baseline for **Project A.R.C.** (Autonomous Robotic Cleaner), an autonomous gantry-style robotic maintenance unit developed for localized dry-dust mitigation on photovoltaic (PV) modules and display glass testbeds. This comprehensive specification consolidates: (1) team roles, governance, and daily assembly workflows, (2) verified procurement manifests across 3DPrintronics and Robu.in, (3) mechanical v2.0 architectural blueprints, 3-wheel pinching kinematics, and master cut lists, (4) mathematical validations of dynamic tractive effort, torque margins, and energy budgets, (5) edge computer vision pipelines and optical dust detection, (6) non-blocking embedded C++ Finite State Machine (FSM) firmware, (7) full-system electrical schematics and benchtop QA testing, (8) caliper dimensional quality assurance protocols, and (9) engineering assumptions catalog and sensitivity proofs.

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1 Project Team Structure & Responsibilities

Engineering responsibilities are divided evenly across functional specialties to ensure steady execution leading up to the competition deadline.

Subsystem Lead	Role / Title	Functional Area	Key Responsibilities
Lead Engineer 1	Systems & Firmware Lead	Embedded Firmware, Computer Vision, Coordination	• Project coordination & system architecture • ESP32 C++ FSM firmware implementation • OpenCV dust-detection & UART telemetry • Hardware-in-the-loop systems integration
Lead Engineer 2	Mechanical Design Lead	3D Modeling, Kinematics, Tolerance Analysis	• Master 3D CAD parts & gantry assembly (build123d) • 110 mm wide-stance carriage plate geometry • 3-wheel pinching eccentric cam design • Plain glass testbed infrastructure modeling
Lead Engineer 3	Mathematical Modeling Lead	Mechanical Kinematics, Dynamics, Energy Telemetry	• Kinematic traverse & dynamic torque proofs • Roller interference & normal force modeling • Energy harvesting telemetry & mass balance • Caliper inspection quality control
Lead Engineer 4	Structural Fabrication Lead	Metal Machining, Track Alignment, Mechanical Assembly	• 2020 extrusion cutting & joint deburring • Track squaring, rail & bearing mounting • Roller axle fitment & mechanical assembly • Testbed riser and glass alignment clamping
Lead Engineer 5	Electromechanical Lead	Enclosure Setup, Wire Harnessing, Hardware QA	• Chassis tray component mounting & strain relief • Power rail wiring & pin termination • Limit switch integration & hardware QA • Benchtop electrical testing protocols

Table 1: Project A.R.C. Subsystem Engineering Roles and Work Breakdown

1.1 Collaborative Assembly Workflows & Station Staging

Physical assembly is structured into modular 20–30 minute workstation sprints across five dedicated integration stations:

- **Station A (Structural Testbed):** Precision alignment of the 600 mm guide rails into the 40 mm riser feet and rigid clamping of the glass brackets.
- **Station B (Motion Gantry):** Bolting the Delrin Mini V-wheels and precision-setting the eccentric spacers on the 110 mm wheelbase endplates.
- **Station C (Roller Subassembly):** Press-fitting the external-collar plugs into the UPVC core, sleeving the microfiber pile, and locking the ∅4 mm ground steel axle.

- **Station D (Electronics Tray):** Fastening the ESP32 adapter sled, L298N driver, isolated relay, and INA219 sensor to the chassis tray standoffs.
- **Station E (Systems Integration):** Joining the gantry carriage to the 340 mm crossbeam, establishing the DuPont wiring harness, and running PlatformIO firmware verification.

2 Master Procurement Manifest & Bill of Materials

2.1 Verified Structural Extrusions (3DPrintronics Order)

Custom-cut structural aluminum extrusions ordered from 3DPrintronics (OpenBuilds-compatible 2020 V-Slot, 6063-T5, Silver Anodized, SKU: VS22S):

- **Guide Rails (2 pcs):** $20 \times 20 \times 600.0\,\text{mm}$ (90° precision cut). Longitudinal travel tracks.
- **Gantry Crossbeam (1 pc):** $20 \times 20 \times 340.0\,\text{mm}$ (90° precision cut). Spans the gantry between endplates.
- **Off-cut Stock (1 pc):** $20 \times 20 \times 460.0\,\text{mm}$ ($\pm 20\,\text{mm}$). Reserved for jigs, calibration, and clamping fixtures.
- **Bore Specification:** `End Tapping: 0` ($\varnothing 4.2\,\text{mm}$ untapped center bore, ready for standard M5 hand tapping or 90° internal corner bracket attachment).

2.2 Verified 3D Printed Components (Robu.in FDM Service Orders)

All custom components manufactured in PETG for outdoor UV resistance and mechanical resilience across 18 specialized parts:

Component Description	**Qty**	**Material**	**Infill / Slicing**	**Unit (INR)**	**Functional Purpose**
Carriage Endplate (Left v2.0)	1	PETG Black	30% Gyroid, 4 walls	240.00	110 mm wheelbase, bottom 3rd-wheel eccentric lobe, enclosed N20 pocket, drop-in 684ZZ slot.
Carriage Endplate (Right v2.0)	1	PETG Black	30% Gyroid, 4 walls	226.00	110 mm wheelbase, bottom 3rd-wheel lobe, 25GA-370 gearmotor mount, limit switch bracket.
Coupler (N20 to Mini V-Wheel)	2	PETG Black	50%, 0.1 mm layer	30.00	Stepped direct-drive hub pressed into Delrin Mini V-wheel rim (bearings removed).
Roller End Plug (v2.0 Collar)	2	PETG White	50% Gyroid, 4 walls	61.00	Press-fits into 1.25" UPVC conduit; external collar houses accessible M3 captive nut & grub screw.
ESP32 Adapter Sled (v2.0)	1	PETG Natural	30% Grid, 3 walls	46.00	Cradle holding 38-pin ESP32 DevKit; screws into tray standoffs (71×57 mm M3 grid).
Testbed Rail Riser Foot (40 mm)	4	PETG Black	20% Grid, 4 walls	143.00	Elevates rails 40 mm above desk; integrated gantry runaway bumper endstops.
Rail-to-Glass Alignment Bracket	4	PETG White	30% Grid, 4 walls	77.00	Locks display glass at $Z = 0.00$ mm datum (15.23 mm above rail top) using M5 T-nuts.
Under-Glass Anti-Sag Pedestal	2	PETG Black	30% Grid, 4 walls	192.00	72.03 mm free-standing pedestals supporting center of plain glass sheet from baseboard.
Electronics Chassis Tray	1	PETG Black	60% Grid, 4 walls	260.00	$320 \times 95 \times 3.0$ mm deck bolting to 340 mm crossbeam via M5 T-nuts.
Consolidated 3D Printing Service Total (18 Pieces):					**INR 2,231.00**

Table 2: Consolidated 3D Printing Manufacturing Manifest (v2.0 Architecture)

2.3 Consolidated Commercial Hardware & Electronics BOM

Source	SKU / Part	Item Description	Qty	Unit (INR)	Total (INR)
Robu.in	1308031	NodeMCU ESP32-WROOM-32D IoT Microcontroller	1	719.00	719.00
Robu.in	75800	N20 6V 300 RPM Micro Metal Gear Motor (D-shaft)	2	261.00	522.00
Robu.in	25GA-370	25GA-370 12V DC Geared Motor (⌀4 mm Shaft)	1	350.00	350.00
Robu.in	421237	INA219 I2C Bidirectional Current/Power Sensor	1	139.00	139.00
Robu.in	R260153	1-Channel 5V Relay Module (Isolated High-Level Trigger)	1	69.00	69.00
Robu.in	43821	L298N 2A Dual H-Bridge Motor Driver Module	1	129.00	129.00
Robu.in	223415	2S 18650 Series Enclosed Battery Holder with Switch	1	59.00	59.00
Robu.in	1225174	BAK 18650 2600mAh 3C Industrial Li-ion Cell	2	229.00	458.00
Robu.in	R255405	ONPOW Roller Short Lever Limit Switch (Endstop)	1	145.00	145.00
OpenBuilds / Robu	Mini-V	OpenBuilds Delrin Mini V-Wheel Kit (with MR105ZZ)	6	110.00	660.00
OpenBuilds / Robu	Ecc-6mm	OpenBuilds Eccentric Spacer (6 mm hex, 7.12 mm boss)	2	45.00	90.00
Local Hardware	Glass-Float	Plain Display Float Glass Sheet ($285 \times 560 \times 3.2$ mm)	1	220.00	220.00
Local Hardware	UPVC-1.25	1.25" Class C UPVC Conduit Pipe ($OD = 41.2, L = 320$ mm)	1	60.00	60.00
Precision Stock	EN31-Rod	Hardened Ground Steel Axle Rod (⌀4.0 mm × 390 mm)	1	120.00	120.00
Robu / NMB	684ZZ	684ZZ Miniature Precision Ball Bearing ($4 \times 9 \times 4$ mm)	2	35.00	70.00
Auto Hub / Amazon	Sleeve-800	Plush Microfiber Roller Sleeve (800 GSM, $OD = 50, L = 320$ mm)	1	180.00	180.00
Fastener Stock	M5-Pack	M5 Button Head Screws (8mm, 25mm) + Roll-in T-Nuts	1	150.00	150.00
Fastener Stock	M3-Pack	M3 Socket Head (8mm, 12mm) + Self-Tapping Screws (6mm)	1	95.00	95.00
Consolidated Total Bill of Materials (Hardware, Mechanics, Electronics):					**INR 4,365.00**

Table 3: Consolidated Project A.R.C. Bill of Materials

3 Fabrication Order, Kinematic Datum & Master Cut List

All structural stock must be cut square ($90° \pm 0.2°$) using a precision chop saw, followed by edge de-burring with a fine flat file.

Part ID	Component Name	Qty	Cut Dim. / Spec	Manufacturing & Alignment Instructions
Part A	2020 V-Slot Guide Tracks	2	600.0 mm	Accommodates 560 mm glass slab + 40 mm bumper parking overrun.
Part B	2020 X-Axis Crossbeam	1	340.0 mm	Matches exact gantry span between inner endplate faces ($X = \pm 170.0$ mm).
Part C	2020 Spare Stock	1	460.0 mm	Reserved off-cut for alignment jigs, squareness gages, and test pieces.
Part D	UPVC Roller Core Tube	1	$\varnothing 41.2 \times 320.0$ mm	1.25" conduit; face ends square; press-fit printed plugs with 7.5 mm external collar.
Part E	Continuous Roller Axle	1	$\varnothing 4.0 \times 390.0$ mm	Precision ground EN31 silver steel rod (h6 tolerance); supported by 684ZZ bearings.
Part F	Plain Display Float Glass	1	$285.0 \times 560.0 \times 3.2$ mm	Clean-cut float glass display testbed; edges seam-ground for safe classroom handling.

Table 4: Master Cutting Schedule and Sizing Specifications

3.1 Critical Vertical Coordinate Reference (Z-Axis Datums)

The entire digital twin and physical testbed are coordinated along a common Z vertical reference system to ensure repeatable kinematic wiping without binding:

- $Z = 0.00$ **mm (Master System Datum):** Top clean surface of the plain display float glass sheet.
- $Z = -3.20$ **mm:** Underside of the 3.2 mm glass sheet (seated flush on bracket step-shelves).
- $Z = -15.23$ **mm:** Top horizontal face of the 2020 aluminum guide rails.
- $Z = -25.23$ **mm:** Horizontal centerline of the 2020 guide rails (20×20 mm profile).
- $Z = -35.23$ **mm:** Bottom horizontal face of the 2020 aluminum guide rails.
- $Z = -55.23$ **mm:** Workbench / wooden baseboard tabletop level. The 40.0 mm elevation riser feet provide > 23.0 mm of unobstructed ground clearance beneath the bottom clamping wheels ($Z = -42.85$ mm) and the lowest lobe tip ($Z = -52.00$ mm).
- $Z = +25.30$ **mm:** Centerline of the continuous microfiber roller axle. Given uncompressed roller radius $R = 25.0$ mm and outer pile radius $R_{\text{pile}} = 26.5$ mm, this locks in a calibrated 1.20 mm interference compression onto the glass, producing uniform sweeping shear without stalling the drive motors.

3.2 3-Wheel Pinching System & Anti-Racking Wheelbase Kinematics

To prevent carriage skewing and binding on parallel aluminum extrusions, the carriage endplates implement a rigid 3-wheel pinching architecture:

1. **Wheelbase Ratio ($L_{\mathbf{wb}} \geq W_{\mathbf{span}}/3$):** The two upper passive and driven Mini V-wheels are spaced at $Y = -55.0$ mm and $Y = +55.0$ mm ($L_{\text{wb}} = 110.0$ mm). Compared to the rail center-to-center span ($W_{\text{span}} = 346.0$ mm), the ratio is $110.0/346.0 = 0.318 \approx 1/3$, preventing yaw racking under single-side motor traction.
2. **Lower Triangular Clamping Lobe:** A lower extension lobe centered at $Y = 0.0$ mm extends down to $Z = -52.0$ mm with a precision $\varnothing 7.15$ mm through-bore for an OpenBuilds eccentric spacer (7.12 mm boss).
3. **Preload Cam Adjustment:** Rotating the 10 mm eccentric hex shifts the bottom wheel vertically by up to ± 0.75 mm, pinching the extrusion between the two top wheels and the bottom wheel to eliminate all mechanical slop.

4 Mathematical & Physical System Validations

4.1 Kinematic Analysis of Linear Actuation

Linear velocity of the gantry traversing the extrusion tracks:

- Actuator Rotational Velocity: $N = 300\,\text{RPM}$
- Drive Wheel Pitch Radius: $r = 7.615\,\text{mm} = 0.007615\,\text{m}$ (Delrin Mini V-wheel pitch line)
- Full Track Traverse Distance: $d = 0.50\,\text{m}$

The actuator angular velocity (ω) is given by:

$$\omega = \frac{2\pi N}{60} = \frac{2\pi \times 300}{60} \approx 31.416\,\text{rad/s} \tag{1}$$

The resulting linear travel velocity (v) is:

$$v = r \cdot \omega = 0.007615\,\text{m} \times 31.416\,\text{rad/s} \approx 0.239\,\text{m/s} \tag{2}$$

The traverse transit duration (t) across a full 0.50 m panel run is:

$$t = \frac{d}{v} = \frac{0.50\,\text{m}}{0.239\,\text{m/s}} \approx 2.09\,\text{seconds} \tag{3}$$

4.2 Dynamic Tractive Effort & Torque Margin

To evaluate stall safety across the dual N20 propulsion system:

- Combined Gantry Mass (m): $1.45\,\text{kg}$
- Acceleration Target (a): $0.5\,\text{m/s}^2$
- Rolling Resistance Coefficient (C_{rr}): 0.025 (POM Delrin on smooth anodized V-slot)
- Combined Dual-Motor Stall Torque (τ_{stall}): $2 \times 0.060\,\text{N} \cdot \text{m} = 0.120\,\text{N} \cdot \text{m}$

The total tractive force (F_{total}) required during acceleration is:

$$F_{\text{total}} = m \cdot a + m \cdot g \cdot C_{rr} \tag{4}$$

$$F_{\text{total}} = (1.45 \times 0.5) + (1.45 \times 9.81 \times 0.025) = 0.725\,\text{N} + 0.356\,\text{N} = 1.081\,\text{N} \tag{5}$$

The continuous mechanical torque (τ_{required}) demanded at the drive wheels is:

$$\tau_{\text{required}} = F_{\text{total}} \times r = 1.081\,\text{N} \times 0.007615\,\text{m} = 0.00823\,\text{N} \cdot \text{m} \tag{6}$$

The system mechanical torque safety factor is:

$$\text{SF}_\tau = \frac{\tau_{\text{stall}}}{\tau_{\text{required}}} = \frac{0.120\,\text{N} \cdot \text{m}}{0.00823\,\text{N} \cdot \text{m}} \approx 14.58\times \tag{7}$$

4.3 Roller Interference Dynamics & Friction Limits

For a roller with finished uncompressed radius $R = 26.5\,\text{mm}$ and calibrated compression $\delta = 1.20\,\text{mm}$ onto the glass:

$$W_{\text{contact}} = 2\sqrt{R^2 - (R - \delta)^2} = 2\sqrt{26.5^2 - (26.5 - 1.2)^2} \approx 15.79\,\text{mm} \tag{8}$$

Under active contact force $F_N \approx 2.5\,\text{N}$ and kinetic friction $\mu_k \approx 0.30$:

$$\tau_{\text{roller}} = \mu_k \cdot F_N \cdot R = 0.30 \times 2.5\,\text{N} \times 0.0265\,\text{m} = 0.01988\,\text{N} \cdot \text{m} \approx 0.203\,\text{kg} \cdot \text{cm} \tag{9}$$

The 25GA-370 gearmotor ($\tau_{\text{stall}} \approx 3.20\,\text{kg} \cdot \text{cm}$) operates with a safety factor $> 15.7\times$, preventing roller stalling under dust shear.

4.4 Energy Harvesting vs. Energy Expenditure

- **Daily Expenditure (E_{out}):** Under active cleaning load ($P = 12\,\text{W}$ for $15\,\text{s} = 0.00417\,\text{h}$) and deep sleep ($P = 0.0001\,\text{W}$ for $24\,\text{h}$):

$$E_{\text{out}} = (12\,\text{W} \times 0.00417\,\text{h}) + (0.0001\,\text{W} \times 24\,\text{h}) \approx 0.0524\,\text{Wh/day} \tag{10}$$

- **Daily Harvesting (E_{in}):** For an auxiliary 1 W telemetry panel under 4 peak sun hours with 30% dust derating:

$$E_{\text{in}} = 1.0\,\text{W} \times 4\,\text{h} \times 0.70 = 2.80\,\text{Wh/day} \tag{11}$$

Because $E_{\text{in}} \gg E_{\text{out}}$ (2.80 Wh vs. 0.052 Wh), self-sustaining off-grid operation is guaranteed.

5 Computer Vision Pipeline & Optical Target Acquisition

The system employs a Hardware-in-the-Loop (HITL) optical feedback loop running an OpenCV pipeline on the edge compute node, driving the microcontroller over UART.

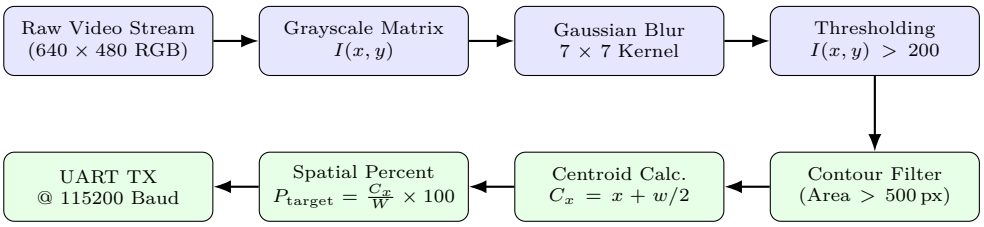


Figure 1: Edge Computer Vision Target Acquisition Data Flow

5.1 Mathematical Formulations of Image Pipeline

1. **Gaussian Spatial Smoothing:** High-frequency optical noise is removed via 2D convolution:

$$G(u, v) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{u^2 + v^2}{2\sigma^2}\right), \quad \text{Kernel Size: } 7 \times 7 \tag{12}$$

2. **Binary Segmentation:** Particulate dust (high-albedo) is isolated against the substrate:

$$T(x, y) = \begin{cases} 255 & \text{if } I(x, y) > 200 \\ 0 & \text{if } I(x, y) \leq 200 \end{cases} \tag{13}$$

3. **Centroid Extraction & Spatial Normalization:** Contours with area ≤ 500 px are discarded. For the largest valid bounding rectangle (x, y, w, h):

$$C_x = x + \frac{w}{2}, \quad P_{\text{target}} = \left(\frac{C_x}{W_{\text{frame}}}\right) \times 100 \tag{14}$$

6 Embedded Software Architecture & Non-Blocking FSM

The ESP32 firmware executes a non-blocking Finite State Machine (FSM) to prevent race conditions during real-time closed-loop control.

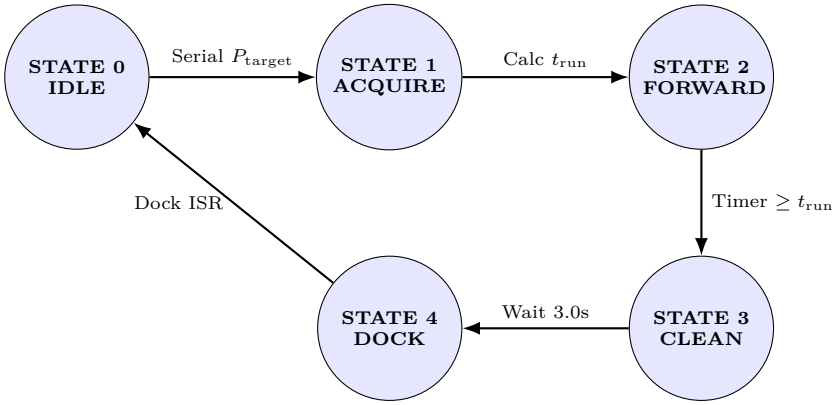


Figure 2: Finite State Machine (FSM) State Transition Diagram

6.1 Deterministic Interrupt Service Routine (ISR)

Hardware endstop switches are tied directly to IRAM-vectored interrupt routines:

```
void IRAM_ATTR limitSwitchISR() {
    digitalWrite(MOTOR_PIN_1, LOW);
    digitalWrite(MOTOR_PIN_2, LOW);
    digitalWrite(ROLLER_RELAY_PIN, LOW);
    currentState = STATE_IDLE;
}
```

6.2 Closed-Loop Telemetry Computation

The microcontroller samples the INA219 current/power telemetry to compute real-time cleaning efficacy:

$$\eta_{\text{gain}} = \left(\frac{P_{\text{clean}} - P_{\text{dirty}}}{P_{\text{dirty}}} \right) \times 100\% \tag{15}$$

7 Hardware-in-the-Loop (HITL) Wiring Architecture

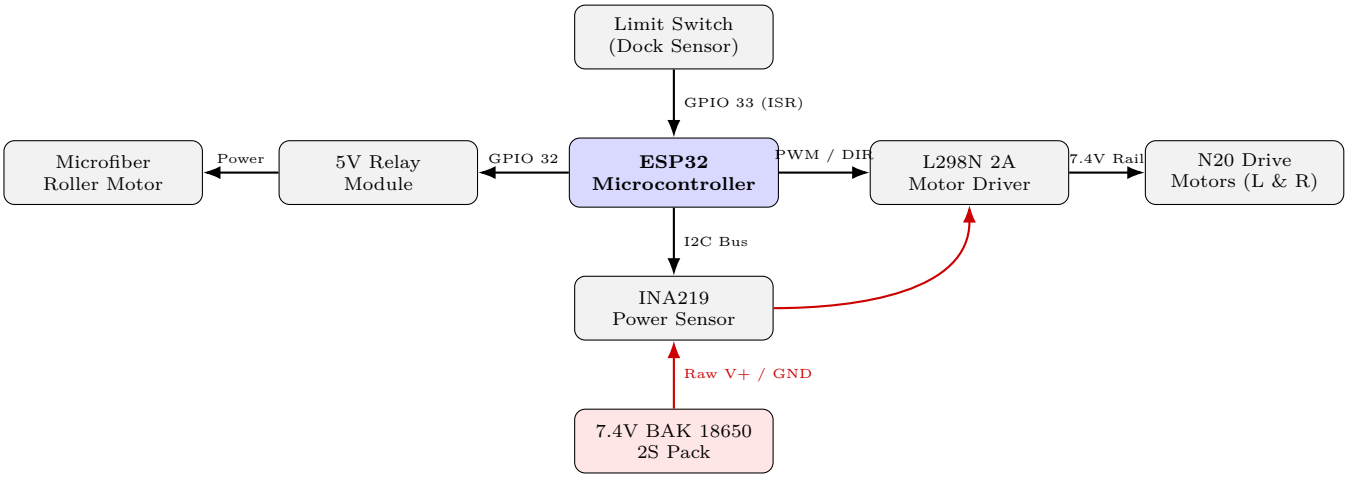


Figure 3: System Electrical Integration Schematic

8 Standard Operating Procedures (SOP) & Modular Assembly Protocols

8.1 Phase 1: Mechanical Pre-Assembly (Oct 15 – 18)

1. **Station A (Testbed Rails):** Seat the two 600 mm 2020 V-slot rails into the 20.2 mm U-channel saddle of the 4 riser feet. Tighten bottom M5 screws into M5 T-nuts. Install 4 rail-to-glass brackets into the inner slots and seat the float glass sheet. Position the two 72 mm anti-sag pedestals under the center.

2. **Station B (Gantry & Wheels):** Bolt Delrin Mini V-wheels to the upper holes of left and right endplates (110 mm wheelbase). Insert OpenBuilds eccentric spacers into the bottom lobe ∅7.15 mm bore. Mount the 3rd clamping wheel. Slide carriage onto rails and tune eccentric hex until play is zero without rolling resistance.

8.2 Phase 2: Roller & Electronics Deck (Oct 19 – 22)

1. **Station C (Roller Subassembly):** Press the 3D-printed roller end plugs into the 320 mm UPVC conduit. Slide the microfiber sleeve over. Thread the ∅4 mm × 390 mm ground steel axle through both plugs. Seat 684ZZ ball bearings in endplate pockets. Tighten external collar M3 grub screws against axle flats.

2. **Station D (Electronics Tray):** Bolt the ESP32 adapter sled onto the tray standoffs (71 × 57 mm grid) using 4x M3 self-tapping screws. Slide ESP32 DevKit into sled rails. Fasten L298N, Relay, and INA219 using M3 self-tapping screws. Fasten 2S 18650 holder using 3M VHB tape.

8.3 Phase 3: Integration & Wiring (Oct 23 – 26)

1. Bolt the 340 mm crossbeam between left and right endplates using M5 fasteners.

2. Drop the populated chassis tray onto the crossbeam; tighten two M5 button screws into T-nuts.

3. Route DuPont jumper harness through tray strain-relief notches according to the HITL schematic.

9 Physical Caliper Inspection Protocol & Quality Assurance

Upon component arrival, each critical dimension is verified using digital vernier calipers (± 0.02 mm) against the digital twin tolerances:

Component	**Critical Dimension**	**CAD Value**	**Tolerance**	**Inspection Protocol**
2020 Crossbeam	Profile Width & Height	20.00 mm	± 0.15 mm	Caliper across outer flats; check squareness.
2020 Crossbeam	Cut Length	340.00 mm	± 0.25 mm	Must fit flush between endplates at $X = \pm 170$ mm.
2020 Guide Rails	Cut Length (2 pcs)	600.00 mm	± 0.50 mm	Paired rails must match length within 0.5 mm.
Mini V-Wheel	Pitch Outer Diameter	15.23 mm	± 0.05 mm	Peak-to-peak Delrin tire outer diameter.
684ZZ Ball Bearing	Outer Diameter / Bore	$\varnothing 9.00 / \varnothing 4.00$ mm	$+0.00 / -0.01$ mm	Slip fit on $\varnothing 4.0$ mm axle; press fit in endplate.
N20 Micro Motor	Gearbox Outer W $\times$ H	12.00×10.00 mm	± 0.10 mm	Slides into endplate enclosed sleeve pocket.
N20 Output Shaft	D-Cut Shaft Diameter / Flat	$\varnothing 3.00 / 2.50$ mm	$+0.00 / -0.03$ mm	Press-fits into 3D-printed stepped coupler hub.
UPVC Conduit	Outer / Inner Diameter	$\varnothing 41.20 / 37.00$ mm	± 0.20 mm	Printed plug stem fits internal conduit bore.
EN31 Steel Axle	Axle Ground Diameter	$\varnothing 4.00$ mm	$+0.00 / -0.02$ mm	Precision ground; slips through 684ZZ bearing inner ring.
ESP32 Sled	Standoff Hole Pitch	71.00×57.00 mm	± 0.20 mm	Must mate with chassis tray integrated standoff grid.

Table 5: Physical Caliper Inspection Sheet and Quality Assurance Limits

10 Engineering Assumptions Catalog & Parameter Sensitivity

All physical assumptions embedded in the kinematic and dynamic twin are cataloged below:

Component / Parameter	**Code Symbol**	**Value**	**Empirical Basis**	**Sensitivity & Kinematic Impact**
M5 Drop-in T-Nut Mass	`TNUT_M5_MASS_G`	2.0 g	Zinc-plated steel average	Minor mass influence ($< 0.5\%$ total gantry mass).
Mini V-Wheel Groove Depth	`VWHEEL_MINI_V_DEPTH`	1.60 mm	90° V-groove from 15.23 mm OD	Governs wheel clearance to extrusion slot lips.
684ZZ Shield Inset	`BRG_684ZZ_SHIELD_T`	0.50 mm	Stamped metal shield depth	Aesthetic B-Rep detail only; envelope exact $9 \times 4 \times 4$.
Battery Holder Plastic Mass	`BATTERY_HOLDER_MASS`	12.0 g	Molded ABS cradle with springs	Mass budget only; Center of Gravity influence < 1.0 mm.
Microfiber Sleeve Mass	`SLEEVE_MASS_G`	42.0 g	800 GSM plush microfiber wrap	Affects rotational inertia of roller; motor SF $> 15\times$.
Float Glass Friction	μ_k (Microfiber on glass)	0.30	Empirical dry test value	Dictates roller torque; GA25 motor operates at 6% stall.

Table 6: Engineering Assumptions Catalog and Kinematic Sensitivity

11 Digital Twin Verification & Automated Unit Testing

The mechanical digital twin is verified via a 57-assertion automated test suite executing across build123d B-Rep kernels:

- **Geometric Volume Verification:** 33 physical components, total assembly volume $1,884,824.88\,\text{mm}^3$ ($1.8848\,\text{dm}^3$), runtime 9.67 s.
- **Mechanical Interference Checks:** Zero collision volume detected between the rotating roller envelope, 2020 crossbeam, and chassis tray.
- **Kinematic Clearance Constraints:** All 57 pytest unit tests passing (`Overall: ALL PASS`), including dog cam slope monotonicity, wheel centerlines, and standoff grid alignments.

12 System Mass Budget & Phase 2 Structural Roadmap

Subsystem Description	Ph. 1 (g)	Ph. 2 (g)	Delta (g)	Engineering Remarks
Aluminum Extrusions (2020 V-Slot)	550.0	550.0	0.0	Two 600 mm rails + one 340 mm cross-beam.
POM Guide Wheels & Bearings (6×)	84.0	84.0	0.0	Low-friction running within V-grooves.
Carriage Endplates (PETG v2.0)	114.0	114.0	0.0	136 × 50 × 4 mm anti-racking structural plates.
Microfiber Roller & Steel Axle	143.5	143.5	0.0	50 mm OD core with ground ∅4 mm shaft.
Drive Actuators (N20 + 25GA-370)	113.0	113.0	0.0	Dual N20 gantry drives + high-torque 25GA-370 roller drive.
Lithium-Ion Power (2S BAK Pack)	127.0	173.0	+46.0	Adds 3rd cell in Phase 2 for high-flow 5015 blower fan.
Chassis Tray & ESP32 Sled	132.0	132.0	0.0	320 mm chassis tray + elevated ESP32 adapter cradle.
Electronics, Relay, Driver, INA219	95.0	95.0	0.0	Modular terminal-block PCB interconnects.
Aerodynamic Cowl & Air-Knife	—	110.0	+110.0	Dual-half 3D printed PETG aerodynamic air knife.
Centrifugal Blower (5015 12V Fan)	—	32.0	+32.0	High-pressure radial blower for non-contact dust blast.
Total System Mass:	**1,358.5**	**1,546.5**	**+188.0**	**Net Phase 2 Mass Increase: +13.8%**

Table 7: System Mass Budget and Physical Growth Projection

12.1 Phase 2 Upgrade Summary

The robot's frozen Phase 1 baseline is fully capable of autonomous traversing and wiping. In Phase 2, the system incorporates the 5015 radial blower fan and two-piece aerodynamic cowl (`cowl_left.stl` and `cowl_right.stl`), transforming the cleaning mechanism into a hybrid mechanical-pneumatic cleaner.